

Mixed Strategies and Nash Equilibria

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Beyond Pure Strategies

- So far, strategies have been **pure**: choose one action deterministically.
- Today we allow players to **randomise** by playing a **mixed strategy**.
- This requires cardinal payoffs, connects back to expected utility, and completes the Nash equilibrium story.

Associated Reading

Game Theory, Sections 6.1–6.3

Why Randomise?

In some games, any deterministic choice can be exploited.

Consider **Matching Pennies**:

	Heads	Tails
Heads	$(+1, -1)$	$(-1, +1)$
Tails	$(-1, +1)$	$(+1, -1)$

If Row plays Heads for certain, Column plays Tails. If Row plays Tails for certain, Column plays Heads. There is **no pure strategy Nash equilibrium**.

The Solution: Mix

If Row plays Heads with probability p and Tails with probability $1 - p$:

Column's expected payoff from Heads: $-p + (1 - p) = 1 - 2p$

Column's expected payoff from Tails: $p - (1 - p) = 2p - 1$

Column is indifferent when $1 - 2p = 2p - 1$, i.e., $p = 1/2$.

At $p = 1/2$, Column cannot exploit Row's strategy. Any mix gives the same expected payoff. The **mixed strategy Nash equilibrium** is for both players to mix 50/50.

Mixed Strategy Nash Equilibrium

A **mixed strategy** assigns a probability to each pure strategy.

A **mixed strategy Nash equilibrium** is a profile of (possibly mixed) strategies such that no player can improve their *expected* payoff by unilaterally deviating.

Nash's Theorem (1950): Every finite game has at least one Nash equilibrium in pure or mixed strategies.

The Indifference Principle

In a mixed strategy Nash equilibrium, a player who mixes must be **indifferent** between the strategies they mix over.

Why? If one strategy had higher expected payoff than another, the player would shift all probability to that strategy (pure strategy, not a mix).

This gives us a way to find mixed equilibria: **set the opponent indifferent**.

Battle of the Sexes Revisited

	Opera	Football
Opera	(2, 1)	(0, 0)
Football	(0, 0)	(1, 2)

We found two pure strategy Nash equilibria: (Opera, Opera) and (Football, Football).

Is there also a mixed equilibrium?

Finding the Mixed Equilibrium

Let Row play Opera with probability p .

Column is indifferent when:

$$\begin{aligned}1 \cdot p + 0 \cdot (1 - p) &= 0 \cdot p + 2 \cdot (1 - p) \\ p &= 2 - 2p \implies p = \frac{2}{3}\end{aligned}$$

Let Column play Opera with probability q .

Row is indifferent when:

$$\begin{aligned}2q + 0(1 - q) &= 0 \cdot q + 1 \cdot (1 - q) \\ 2q &= 1 - q \implies q = \frac{1}{3}\end{aligned}$$

Mixed equilibrium: Row plays Opera with probability $\frac{2}{3}$; Column plays

Expected Payoffs in the Mixed Equilibrium

Row's expected payoff:

$$2 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3} = \frac{2}{3}$$

Column's expected payoff:

$$1 \cdot \frac{2}{3} + 0 \cdot \frac{1}{3} = \frac{2}{3}$$

Both do **worse** in the mixed equilibrium ($\frac{2}{3}$) than in either pure equilibrium (Row gets 2 or 1; Column gets 1 or 2).

The mixed equilibrium is the outcome when players can't coordinate on which pure equilibrium to play.

Cardinal Payoffs and Zero-Sum Games

Why Cardinal Payoffs Matter Now

For pure strategy analysis, ordinal payoffs suffice. We just need to know which outcomes are ranked higher.

For mixed strategies, we need **cardinal** payoffs. The actual numbers matter, not just the order.

This is because mixed strategy analysis is expected utility calculation. The magnitudes of payoffs affect the optimal mixing probabilities.

A **zero-sum game** is one where every gain for one player is an equal loss for another:

$$u_1(s) + u_2(s) = 0 \text{ for all strategy profiles } s$$

Matching Pennies is zero-sum. Poker is (approximately) zero-sum.

In zero-sum games, the interests of players are **perfectly opposed**.

The Minimax Theorem

For two-player zero-sum games:

von Neumann's Minimax Theorem (1928): *There is always a mixed strategy profile where each player **maximises their minimum guaranteed payoff** (the minimax strategy).*

The minimax strategy and the Nash equilibrium coincide in zero-sum games.

This was the first major theorem of game theory, proved before Nash's more general result.

Minimax in Matching Pennies

Each player's minimax strategy: mix 50/50. This guarantees an expected payoff of 0.

Against any deterministic opponent, you can't do better on average, but you can't do worse either.

In zero-sum games, the minimax strategy is the uniquely rational strategy.

Interpreting Mixed Strategies

What Does Mixing Mean?

Players don't literally flip coins at the start of a game (usually).

Several interpretations:

- **Deliberate randomisation:** a player genuinely randomises to stay unpredictable (relevant in sports, military strategy).
- **Population interpretation:** the equilibrium describes a *distribution* across a large population of players, not an individual's behaviour.

Mixed strategy equilibria are especially relevant when:

- **Unpredictability has strategic value:** penalty kicks in soccer, play-calling in American football, auditing and compliance.
- **Coordination fails:** the mixed equilibrium is the “default” when no focal pure equilibrium exists.

Worked Example

A Technology Standards Game

Two firms simultaneously choose whether to adopt **Standard A** or **Standard B** for a new product. Each prefers that the other adopts the same standard (compatibility is valuable), but Firm 1 has a slight preference for A and Firm 2 for B.

	Firm 2: A	Firm 2: B
Firm 1: A	(3, 2)	(0, 0)
Firm 1: B	(0, 0)	(2, 3)

Step 1: Find Pure-Strategy NE

(A, A): Firm 1 gets 3, switch to B gives 0. No improvement. Firm 2 gets 2, switch to B gives 0. No improvement. **NE.**

(B, B): Firm 1 gets 2, switch to A gives 0. No improvement. Firm 2 gets 3, switch to A gives 0. No improvement. **NE.**

(A, B) and (B, A): both give (0, 0), and both firms can improve by switching.
Not NE.

Two pure equilibria. Is there also a mixed one?

Step 2: Set Up the Indifference Conditions

Let Firm 1 play A with probability p and Firm 2 play A with probability q .

Make Firm 1 indifferent (Firm 1's payoff from A = payoff from B):

$$3q + 0(1 - q) = 0 \cdot q + 2(1 - q)$$

$$3q = 2 - 2q$$

$$5q = 2 \implies q = \frac{2}{5}$$

Step 2 Continued

Make Firm 2 indifferent (Firm 2's payoff from A = payoff from B):

$$2p + 0(1 - p) = 0 \cdot p + 3(1 - p)$$

$$2p = 3 - 3p$$

$$5p = 3 \implies p = \frac{3}{5}$$

Mixed equilibrium: Firm 1 plays A with probability $\frac{3}{5}$; Firm 2 plays A with probability $\frac{2}{5}$.

Step 3: Expected Payoffs

Firm 1's expected payoff (using either pure strategy, since indifferent):

$$3 \cdot \frac{2}{5} + 0 \cdot \frac{3}{5} = \frac{6}{5} = 1.2$$

Firm 2's expected payoff:

$$2 \cdot \frac{3}{5} + 0 \cdot \frac{2}{5} = \frac{6}{5} = 1.2$$

Both get 1.2 in the mixed equilibrium, worse than *either* pure equilibrium (where someone gets 2 and someone gets 3).

The mixed equilibrium represents the outcome when coordination fails. Each firm leans toward its preferred standard, and sometimes they end up mismatched.

- We'll do some revision, and some more examples of mixed equilibria.